

Seh Na Mellick

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Machine learning, automated microscopy, and evolutionary modeling for understanding and treating bacteria.

EDUCATION

Carnegie Mellon University Pittsburgh, PA, 2024 – Present

Ph.D. Computational Biology, GPA: 3.8 (expected 2029)

- Coursework: Machine Learning (PyTorch) · Computational Structural Biology · Cellular Systems Modeling · Computational Genomics · Modeling Evolution · Advanced Genetics
- Teaching: 02-712 Computational Methods for Biological Modeling and Simulation (2024) · 02-250 Introduction to Computational Biology (2025)

University of Nebraska Omaha Omaha, NE, 2019 – 2024

B.S. Molecular & Biomedical Biology, GPA: 3.9, Minors: Chemistry & Mathematics

- Graduate Coursework: Discrete Event Simulations Modeling · Boolean Network Models
- Undergraduate Coursework: Molecular Genetics · Cellular Biology · Immunology · Molecular Neurobiology · Organic Chemistry · Biochemistry · Machine Learning & Data Mining · Multivariable Calculus · Linear Algebra · Probability & Statistics · Bioinformatics · Deterministic Operations Research · Discrete Mathematics

Central High School Omaha, NE, 2016 – 2019

GPA: 4.8, AP Scholar

RESEARCH EXPERIENCE

Graduate Research Assistant Carja and Bridges Labs @ Carnegie Mellon University, 2024 – Present

- Leveraged automated high-throughput image-based phenotyping to generate large-scale datasets characterizing growth phenotypes of bacterial gene deletion mutant libraries under various conditions.
- Designed image analysis and computer vision pipelines combining vision transformers, quantitative feature extraction, and representation learning to characterize spatiotemporal bacterial growth phenotypes.
- Built models to predict bacterial growth dynamics, classify gene deletions and species, predict antibiotic resistance phenotypes, and identify novel genotype-phenotype relationships.
- Modeled the influence of collective aggregation on evolution of antibiotic resistance in bacterial infections using theoretical population genetics, evolutionary rescue modeling, and stochastic simulations.

AI PhD Resident Bio Design Kitchen @ Google X, The Moonshot Factory, 2026

- Conducted research on early-stage investigation involving AI and LLMs for biological lab automation.
- Developed skills in computer & machine vision, automated experimentation, agentic workflows, woodworking, metalworking, and welding.

Rotation Student Cooper Lab @ Univ. of Pittsburgh, 2024

- Developed experimental and computational models of catheter-associated bacterial infections and performed evolution experiments to validate model predictions of antibiotic resistance evolution *in vitro*.
- Prepared and analyzed Illumina whole-genome sequencing libraries for barcoded samples.

Microbiology Consultant UNeTECH @ Univ. of Nebraska Medical Center, 2021 – 2022

- Designed and conducted microbiology experiments for local businesses and healthcare partners.
- Analyzed and communicated experimental results to industry partners and non-technical stakeholders.

Undergraduate Research Assistant Rowen Lab @ Univ. of Nebraska Omaha, 2020 – 2024

- Performed molecular cloning and microbiology experiments to characterize the role of the putative transcription factor PA5189 in *Pseudomonas aeruginosa* antibiotic resistance and biofilm formation.
- Analyzed RNA-seq and whole-genome sequencing data to identify mutations in strains expressing resistance to a novel antimicrobial peptide.

Research Assistant Soukup Lab @ Creighton University, 2018 – 2019

- Performed molecular cloning and microbiology experiments to assess *glmS* riboswitch ligand analogs for potential use as antibacterial agents in *Staphylococcus aureus* and *Bacillus subtilis*.
- Generated and prepared crystals for X-ray crystallography to determine mRNA riboswitch 3D structures.

PRESENTATIONS & PUBLICATIONS

1. **Mellick, S.**, Derringer, J., Boyes, D., Croteau, G., Burke, M., Gifford, S., Stark, D., Mike, L., Turecki, S., Carja, O., Mikheyeva-Bridges, I., Bridges, D. (2026). Genome-scale label-free imaging reveals cellular physiology encoded in bacterial collective architecture. *bioRxiv* preprint. doi.org/10.64898/2026.08.30.748126.

2. **Mellick, S.**, Rowen, D. (2024). Characterizing the role of PA5189 in *Pseudomonas aeruginosa* deletion and overexpression mutants. *Undergraduate Honors Thesis*, University of Nebraska Omaha.
3. **Mellick, S.**, Rowen, D. (2023). Role of the PA5189 regulon in *Pseudomonas aeruginosa* antibiotic resistance and biofilm formation. *Conference talk*, Nebraska Academy of Sciences, Lincoln, NE.

SELECTED COURSE PROJECTS

Predicting Fitness Effects of Synonymous Mutations *10-710 Computational Genomics, CMU, 2025*

Curated a dataset integrating genomic, evolutionary, kinetic, and structural features of synonymous mutations in *S. cerevisiae* and built interpretable models predicting fitness effects of synonymous mutations.

Probing Chirality in E(3)-Equivariant Boltzmann Generators *MSCBIO 2030 Comp. Structural Biology, Pitt, 2025*

Demonstrated that SO(3)-equivariant Boltzmann generators are necessary to capture chirality in molecular structures and latent representations, with implications for using traditional E(3)-equivariant models for modeling biomolecular systems and drug design.

Adversarial Robustness in ImageNet *10-701 Machine Learning, CMU, 2024*

Studied transferability of localized adversarial perturbations across CNN and Vision Transformer architectures using FGSM, PGD, and Carlini-Wagner adversarial techniques combined with segmentation and bounding-box detection to localize perturbations to specific image regions.

AWARDS & HONORS

Research Awards: Best Poster, CMU Computational Biology Symposium (2026) · 2023 UNO FUSE Grant · 2022 NASA Nebraska Space Grant Fellowship.

Academic Honors: UNO Honors Distinguished Scholar · Omaha Hearing School Scholarship (academic achievement as Deaf/Hard-of-Hearing student) · UNO Chancellor's Scholarship

Music Performance: Nebraska Wind Symphony Memorial Concerto Competition Scholarship (2021) · recipient of five merit-based scholarships for music performance at UNO School of Music

SKILLS

Machine Learning & AI: PyTorch · scikit-learn · vision transformers · CNNs · LLM systems & fine-tuning · LLM agents (LangGraph)

Programming: Python · Julia · R · Git · Bash · Docker/Singularity · HTML, CSS, & JS

Concepts: statistical learning · active learning · HPC · networked systems · computer & machine vision · deterministic & stochastic modeling

Biology: microbial cloning & molecular genetics (PCR, qRT-PCR, plasmid construction, transformation, conjugation) · experimental microbiology (MIC & biofilm assays, microbial imaging) · experimental evolution · mammalian cell culture · bioinformatics (WGS, bulk RNA-Seq, barcode sequencing, phylogenetics) · automated microscopy · liquid handling robotics · brightfield & confocal microscopy image analysis pipelines

Languages: English (native) · French (intermediate) · American Sign Language (novice)

MEMBERSHIPS & ACTIVITIES

Senator: CPCB Graduate Student Assembly (2024–Present)

Member: American Society for Microbiology, (2024–Present) · UNO Honors Program (2019–2024)

Bassist: CMU Indian Music Ensemble (2024–Present) · UNO Maverick Machine (2023–2024)

Principal Oboist: UNO Symphonic Wind Ensemble (2019–2024) · UNO Heartland Philharmonic Orchestra (2019–2023) · UNO University Band (2021–2023)